PRABHAT DUBEY

Senior Software Engineer | AI Infrastructure | Distributed Systems | LLM/GenAI | Data Protection
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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience building high-throughput distributed systems, AI/ML infrastructure, and cloud-native platforms at scale. Architected enterprise-grade solutions processing petabyte-scale data across AWS and Azure ecosystems. Currently building LLM-powered intelligent systems with RAG pipelines and AI agents for enterprise automation. Inventor on 14 granted US patents in cloud storage and virtualization. Proven track record of shipping zero-downtime features, driving cross-functional engineering teams, and integrating AI capabilities into production systems. Currently pursuing M.Tech in AI/ML from BITS Pilani.

TECHNICAL SKILLS

Languages: C++, Java, Python, C#, Go, TypeScript
AI / ML / LLM: PyTorch, LangChain, HuggingFace, RAG, Fine-tuning, Prompt Engineering
GenAI Stack: OpenAI API, Pinecone, ChromaDB, AI Agents, Embeddings
Frameworks: Spring Boot, .NET Core, Angular, Node.js, FastAPI
Cloud & Infra: AWS (S3, SageMaker, Lambda), Azure (Blob, OpenAI), Kubernetes, Docker
Databases: PostgreSQL, MongoDB, Redis, SQLite, Vector Databases
System Design: Distributed Systems, Microservices, ML Pipelines, Model Serving
MLOps & DevOps: MLflow, Jenkins, Git, CI/CD, Weights & Biases, REST/gRPC

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Dell Technologies | Bengaluru, India | Apr 2022 – Present

- Architected SQL Server per-file database relocation feature end-to-end (UI to backend across 6 repositories), reducing the original epic delivery estimate by 70% and enabling zero-downtime restore operations.
- Designed and built Native Edge Plugin Service using Go and Java microservices to automate onboarding and discovery of 1,000+ edge devices, improving setup efficiency by 30% and cutting provisioning time from hours to minutes.
- Engineered vProxy discovery daemon for automated VM discovery in Hyper-V and Native Edge environments, reducing manual intervention by 40% and eliminating configuration drift across distributed clusters.
- Led the Hyper-V Backup & Restore module integrating C++, .NET, and Java services for Dell PowerProtect Data Manager (PPDM), achieving 99.9% restore reliability across 50+ enterprise customers.
- Built GHVDM microservice layer (Spring Boot) bridging PPDM with Hyper-V Agent, processing 10K+ metadata operations/day with sub-second latency for data flow and metadata management.
- Developed AI-powered log analysis pipeline using LLM-based classification and RAG to auto-diagnose backup failures from unstructured logs, reducing mean-time-to-resolution by 45% across support workflows.
- Resolved 15+ critical P0/P1 customer escalations under SLA, mentored 5 engineers on system design, debugging, and performance tuning best practices.

Software Engineer II, Dell Technologies | Bengaluru, India | Dec 2019 – Mar 2022

- Developed Object Store backup/restore SDK for Azure Blob and AWS S3 using C++ and Node.js, improving cloud data transfer throughput by 35% through parallel chunked uploads and connection pooling.
- Architected Active Directory Granular Level Restore (AD GLR) system with iSCSI-based live mount, reducing recovery turnaround from hours to minutes for enterprise AD environments.
- Served as Scrum Lead for AD GLR epic; managed sprint planning, backlog refinement, and delivery tracking for a team of 8 across 4 sprints.
- Optimized backup verification pipeline using SQLite and VSS integration, reducing recovery time by 25% through query optimization and batch processing.
- Named inventor on 14 granted US patents covering cloud storage optimization, Office 365 backup, and virtualization recovery techniques.

Software Engineer, Micro Focus (now OpenText) | Bengaluru, India | Aug 2016 – Nov 2019

- Built core modules for Directory and Resource Administrator (DRA) product using C++/.NET backend and Angular frontend, improving feature delivery velocity by 20%.
- Developed enterprise desktop applications using WCF and WinForms for administrative workflows, service communication, and configuration management within the DRA ecosystem.
- Designed WCF-based service layers enabling secure, encrypted communication between management consoles and backend services across distributed Active Directory domains.
- Created standalone Health Check Utility (WinForms) for product validation and diagnostics; recognized at internal hackathon for innovation and adopted org-wide.

KEY PROJECTS

- **Intelligent Log Diagnostics (RAG + LLM):** Built a Retrieval-Augmented Generation pipeline using LangChain, ChromaDB, and GPT-4 to auto-classify and diagnose backup failure logs, replacing manual triage for 10K+ daily incidents.
- **AI-Driven Backup Policy Recommender:** Developed an ML-based system using Python and scikit-learn to analyze historical backup patterns and recommend optimal protection policies, improving storage efficiency by 20%.
- **Native Edge Orchestrator:** Created Go-based microservices for edge node onboarding, discovery, and secure configuration management with zero-touch provisioning.
- **Object Store Cloud Backup:** Architected high-throughput upload/download modules for Azure & AWS with multi-part parallel transfers, supporting petabyte-scale distributed storage.
- **Active Directory GLR:** Built iSCSI-based live mount restore system reducing AD recovery from hours to minutes for enterprises with 100K+ directory objects.
- **Bare Metal Recovery (BMR/SSR):** Designed low-level backup/restore mechanisms in C++ and VSS enabling complete OS state recovery with data consistency guarantees across heterogeneous hardware.
- **Hypervisor Protection (Hyper-V Backup & Restore):** Led the Hyper-V backup and restore module for Dell PowerProtect Data Manager, spanning C++, .NET and Java services. Built the Spring Boot microservice layer that bridges PPDM with the Hyper-V Agent (10K+ metadata operations a day) and a discovery daemon that finds VMs automatically across Hyper-V and Native Edge environments. Delivered 99.9% restore reliability across 50+ enterprise customers.
- **Directory and Resource Administrator (DRA):** Enterprise identity management product for secure, delegated administration of Active Directory environments, built at Micro Focus (now OpenText). Developed core modules on a C++/.NET backend with an Angular frontend, and WCF service layers providing encrypted communication between management consoles and backend services across distributed AD domains. Also built WCF and WinForms desktop applications and a Health Check Utility that won internal hackathon recognition and was adopted org-wide.

PATENTS & INTELLECTUAL PROPERTY

14 US patents granted covering cloud object storage optimization, Office 365 backup strategies, Hyper-V virtualization recovery, incremental data deduplication, and distributed data protection architectures.

US 11,095,745; US 11,829,630; US 12,216,549; US 11,836,046; US 12,105,602; US 11,809,280; US 12,625,999; US 12,032,445; US 11,340,824; US 11,442,819; US 12,026,062; US 11,669,403; US 11,734,124; US 11,442,822. Public records: patents.justia.com

EDUCATION

M.Tech in Artificial Intelligence & Machine Learning, BITS Pilani (WILP) | Pursuing (2026 - Present)
B.Tech in Information Technology, Guru Ghasidas Central University, Bilaspur | CGPA 7.96 / 10 | 2012 - 2016

ACHIEVEMENTS & LEADERSHIP

- Scrum Lead for multiple product releases in Dell's Data Protection suite, coordinating 8-member cross-functional teams.
- Delivered 10+ internal tech talks on backup architecture, cloud integration, system design, and AI/ML integration patterns.
- Championed AI-first development practices within the team, integrating LLM-based tooling into code review and incident response workflows.
- Recognized for resolving critical customer escalations (P0/P1) under tight SLAs with zero recurring incidents.
- Mentored 5+ junior and mid-level engineers on system design, distributed systems debugging, and AI-augmented development.